

Abhinav Shakya

✉ abhinavshakya063@gmail.com | [in linkedin.com/in/abhinav-shakya](https://www.linkedin.com/in/abhinav-shakya) | 📍 Uttar Pradesh, India

EDUCATION

SRM Institute of Science and Technology

Chennai, India

Bachelor of Technology in Artificial Intelligence and Machine Learning

June 2023 – May 2027

- Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Computer Vision, Neural Networks

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, SQL, LaTeX

Machine Learning/AI: TensorFlow, PyTorch, Scikit-learn, Keras, OpenCV, CNNs, RNNs, Transformers, NLP

Web Development: React, Node.js, Express.js, HTML/CSS, RESTful APIs, Full-Stack Development

IoT & Embedded Systems: Arduino, Raspberry Pi, Drone Systems, Autonomous Navigation, Sensor Integration

Databases: MySQL, PostgreSQL, MongoDB, Firebase

Tools & Platforms: Git, GitHub, Docker, Linux, Jupyter, Google Cloud Platform, AWS, Agile/Scrum

Cybersecurity: Intrusion Detection Systems, Network Security, Threat Analysis, Security Protocols

EXPERIENCE

Artificial Intelligence and Machine Learning Intern

June 2024 – July 2024

Extion Infotech

Remote

- Developed machine learning models for real-world applications, improving model accuracy by implementing advanced feature engineering techniques
- Collaborated with cross-functional teams using Agile methodologies to deliver AI solutions on schedule
- Gained hands-on experience in project management, code reviews, and deployment of ML models in production environments
- Participated in technical discussions on model optimization, scalability, and performance tuning

AI & Software Developer

Present

Canopy Puffs

Collaborative Project

- Architecting AI-powered autonomous drone systems for search and rescue operations, medical aid delivery, and precision agriculture
- Implementing computer vision algorithms for object detection, obstacle avoidance, and intelligent path planning
- Building real-time decision-making systems using deep learning models for autonomous navigation in dynamic environments
- Integrating IoT sensors and developing embedded software for drone hardware control and data acquisition

PROJECTS

AI-Powered Intrusion Detection System | *Python, TensorFlow, CNNs, RNNs, Network Security*

- Designed and implemented a deep learning-based intrusion detection system achieving 95%+ accuracy on network traffic classification
- Utilized Convolutional and Recurrent Neural Networks to identify anomalous patterns and potential security threats in real-time
- Preprocessed and analyzed large-scale network datasets, optimizing model performance through hyperparameter tuning

Smart Agriculture: Fruit & Vegetable Health Assessment | *Python, Computer Vision, CNNs, Image Processing*

- Developed CNN-based image classification system to detect diseases and assess health of fruits and vegetables with 92% accuracy
- Implemented data augmentation and transfer learning techniques using pre-trained models (ResNet, VGG) to enhance model generalization
- Created an end-to-end pipeline from image preprocessing to model deployment for real-world agricultural applications

Autonomous Drone Navigation System | *Python, OpenCV, ROS, Path Planning Algorithms, Embedded Systems*

- Building intelligent navigation system for autonomous drones using sensor fusion, SLAM, and reinforcement learning
- Developing computer vision modules for real-time object detection and tracking in GPS-denied environments
- Implementing safety-critical decision-making algorithms for obstacle avoidance and emergency response scenarios

Full-Stack Web Applications & Visualization Tools | *React, Node.js, Express, MongoDB, D3.js*

- Built responsive full-stack web applications with RESTful APIs, user authentication, and real-time data visualization
- Designed interactive dashboards for data analytics using modern JavaScript frameworks and visualization libraries
- Optimized application performance through efficient algorithms, caching strategies, and database query optimization

CERTIFICATIONS

- Career Essentials in Cybersecurity – Microsoft & LinkedIn
- Career Essentials in Generative AI – Microsoft & LinkedIn
- What Is Generative AI? – LinkedIn Learning
- AI for All: From Basics to GenAI Practice – Course Completion
- Artificial Intelligence Primer Certification

KEY STRENGTHS

- **Problem Solving:** Strong foundation in data structures, algorithms, and system design with focus on scalable solutions
- **Research & Innovation:** Self-driven approach to learning cutting-edge AI technologies and applying them to real-world challenges
- **Interdisciplinary Expertise:** Unique combination of AI/ML, IoT, cybersecurity, and embedded systems knowledge
- **Collaboration:** Proven ability to work in cross-functional teams using Agile methodologies and version control best practices